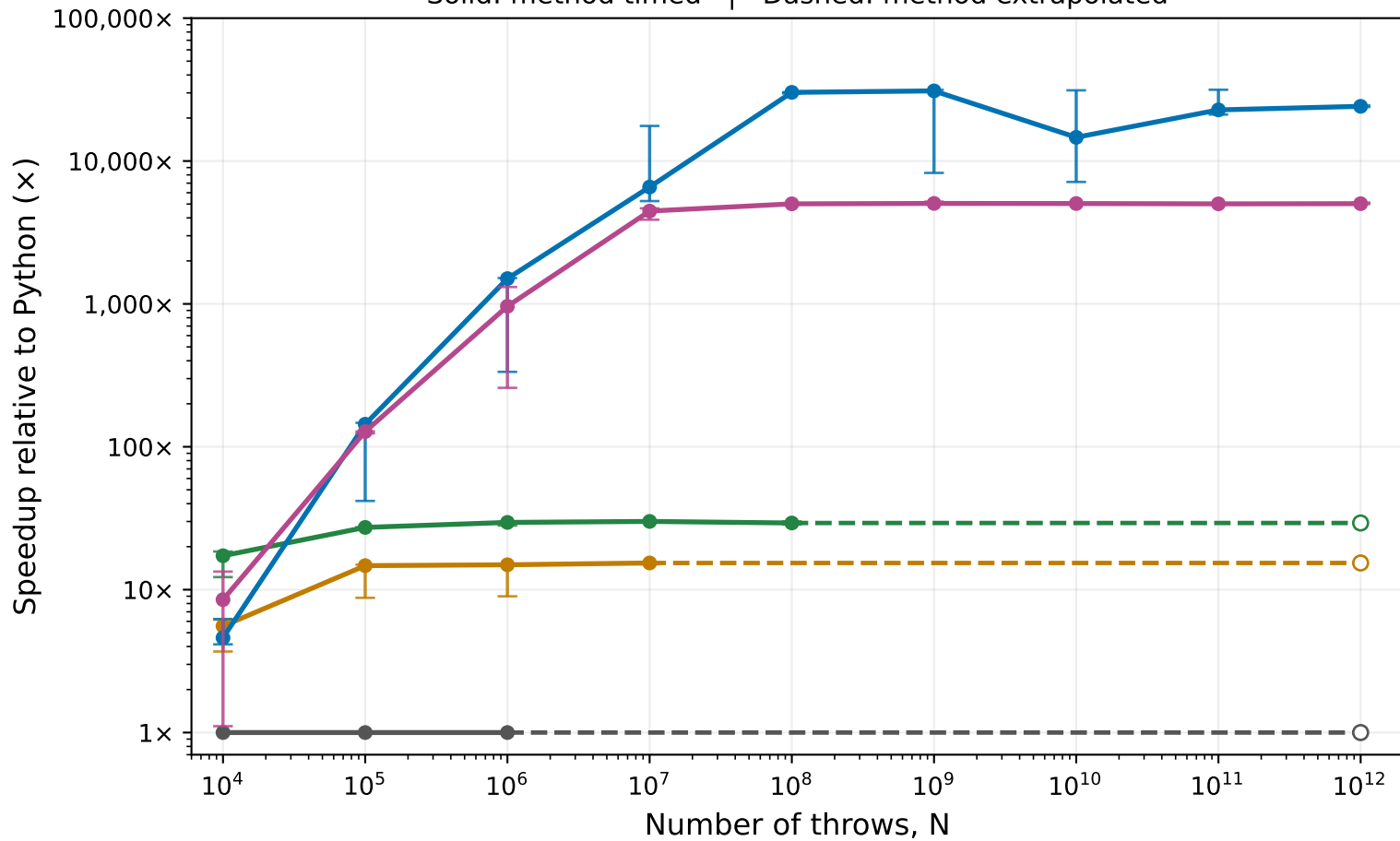


Buffon's needle: speedup from Python to GPU

Molab: 4 CPUs | NVIDIA RTX PRO 6000 Blackwell Server Edition

Solid: method timed | Dashed: method extrapolated



Speedup = Python time / method time. Python uses math.sin; its baseline is extrapolated beyond 10⁶ throws.

Median of 3 runs at every point. Whiskers: min-max method timing range; Python-reference uncertainty excluded.

GPU: shuffled repeat order, 1 s sustained warmup before each run. Batches: NumPy 10⁶, GPU 10⁷. Compilation excluded.

At one trillion throws

Python

1.0×

NumPy · batched

15.4×

extrapolated method runtime

Numba · parallel (4 CPUs)

29.3×

extrapolated method runtime

GPU · float32

24,145×

measured GPU runtime

GPU · float64

5,029×

measured GPU runtime